Lectures

STAT 109: Introductory Biostatistics

Lectures

Below each module: **materials** and **outcomes** — what you should be able to do when you finish that module.

Summer async (2026): Materials are grouped by **module** (0–5 on the course path). Each step is tagged **essential** or **optional** on that path.

Module 0–1 practice quizzes: Quiz 0 (Start Here) · Quiz 0B (Question Ideas) · Quiz A (Def & Notation) · Quiz B (Concept Check) · Synthesis. Later modules use their own quiz banks on Canvas.

Module 0 — Start here

Complete before Module 1. See the course path for the syllabus, reference sheets, and labs.

Step	Material
Read 0	Module 0 Read 0 — welcome, statistical process, three worlds, five projects, Colab & R
Read	Syllabus · Course path · Resources
Quiz	Quiz 0 (Start Here) · Quiz 0B (Question Ideas)
Reference sheets	Methods map · R reference
Lab	Demo lab · Exercise lab
Canvas	Module 0 discussion (introduce yourself)

After Module 0 you should be able to:

- Navigate where readings, labs, and projects live vs what happens in Canvas.
- Open R in Google Colab, save a copy, and run notebook cells.
- Use variables, vectors, logical TRUE/FALSE, and a simple simulation loop.

Module 1 — Testing a Claim

Is this observation surprising?

Course path

Step	Material
Read 0	Module 1 Read 0 — project preview & sample tour
Read	Module 1 Read 1
Quiz A	Def & Notation
Read more	Module 1 Read 2
Quiz B	Concept Check
Demo lab	Module 1 demo · Conditional demo
Exercise lab	Module 1 exercise lab
R reference	R quick reference (Module 0–1 + Project 1)

Step	Material
Milestone	Project 1 milestone · notebook
Synthesis	Module 1 synthesis
Project	Project 1

Read 1 — Probability through the binomial pmf

Essential · after Read 0

- Compute counts, proportions, and percentages; use x, n, $\hat{p}$, and $P(\text{event})$.
- Define sample space, outcome, and event; use the equally likely rule.
- Apply addition, complement, conditional, and multiplication rules; check independence.
- Use permutations and combinations; define binomial process and pmf.

Read 2 — Binomial test in practice

Essential · after Read 1 and Quiz A

- Translate a research question into H_0 and H_a; verify BRP conditions.
- Plot the null binomial distribution; mark observed x; compute and interpret a p-value.
- Carry out the full Project 1 workflow for one- and two-sided alternatives.

Module 2 — Estimating the Truth

Building high-quality estimates from samples.

Course path

Step	Material
Read 0	Module 2 Read 0 — project preview & sample tour
Read	Module 2 Read 1
Quiz A	Def & Notation
Read more	Module 2 Read 2
Quiz B	Concept Check
Demo lab	Module 2 demo
Exercise lab	Module 2 exercise lab
Milestone	Project 2 milestone *(Canvas checkpoint)*
Synthesis	Module 2 synthesis
Project	Project 2

Read 1 — Binomial center/spread through sampling and CIs

Essential · after Read 0

- Use $\mu = np$ and $\sigma = \sqrt{np(1-p)}$ for $X \sim \text{Binomial}(n, p)$.
- Interpret the mean as the **center** and the SD as **typical deviation** of X.
- Identify n and p in applied settings and compute μ and σ.
- Use the **CLT** to approximate a binomial distribution by a normal curve with the same μ and σ.
- Check conditions such as $np \geq 10$ and $n(1-p) \geq 10$ when appropriate.
- State the **Empirical Rule** for normal distributions.
- Sketch a **rejection region** under a normal curve for a one-proportion test and relate the shaded area to the p-value (approximate).

- Define the **sampling distribution** of a statistic.
- Describe **confidence level**, **parameter**, and the endpoints of a confidence interval in context.
- Compute and interpret a confidence interval for a **population proportion** and for a **population mean** (methods from lecture).
- Discuss **bias** and its impact given how a sample was collected.
- Identify ways to reduce bias and increase **precision**.
- Identify a **sampling frame** and carry out **simple random sampling**.
- Classify methods as **convenience**, **SRS**, **cluster**, **systematic**, **stratified**, or **multi-stage**.
- Identify a **sampling frame** for a stated population.
- **Carry out** simple random sampling using a numbered list and random draws.
- **Classify** sampling designs and discuss bias, generalizability, and how to improve the design.

Optional enrichment (not required for Project 2): Type I/II errors · Rubber-band CI analogy

Read 2 — z-scores, toolbox, and one numerical variable

Essential · after Read 1 and Quiz A

- Compute a **z-score** from an observation and the **mean** and **standard deviation** of a reference distribution.
- **Interpret** a z-score in context and use z-scores for **standardized comparisons** between groups or to a reference.
- Find **probabilities** (areas) for the **standard normal** distribution and describe them as proportions of values below (or above) a cutoff.
- Compute a **standardized test statistic** using the **normal approximation to the binomial** and connect it to a **one-proportion** test (often called a **z-test**); in R, use `prop.test()` and read off a **confidence interval** as well.
- Organize analyses by **type of variable(s)** and by **purpose** (e.g. description vs. inference).
- Relate **descriptive** tools to **inferential** choices for **one categorical** variable (e.g. proportion, confidence interval vs. hypothesis test).
- Compute **mean** and **median** by hand and with the **tidyverse** (pipe to `summarize()`).
- Graph one numerical variable with **ggplot2** using a **histogram** and a **dot plot**.
- State the **one-sample *t*-statistic** and use `t.test()` for a **one-sample *t*-test** and **confidence interval** for the **population mean**.

Optional for Project 2: one-sample `t.test()` for hypothesis testing (CI is the project focus).

Legacy lecture pages remain linked from the 15-week index for reference.

Module 3 — Discovering Patterns

Describing a dataset with graphs and summaries.

Course path

Step	Material
Read 0	Module 3 Read 0 — project preview & EDA tour
Read	Module 3 Read 1
Quiz A	Def & Notation
Read more	Module 3 Read 2
Quiz B	Concept Check
Demo lab	Module 3 demo
Exercise lab	Module 3 exercise lab
Milestone	Canvas Module 3 *(dataset + four variables plan)*
Synthesis	Module 3 synthesis
Project	Project 3

Read 1 — Two groups, spread, and boxplots

Essential · after Read 0

- Compute **range**, **quartiles**, **IQR**, and **standard deviation** by hand and **compare** two small datasets.
- Build and interpret **boxplots** as a picture of the **five-number summary**.
- Describe the **Welch two-sample *t*-test** (independent groups, **unequal variances**), its ***t*-statistic**, **degrees of freedom**, **hypotheses**, and **conditions**; run `t.test(y ~ group)` in R.
- Use `group_by()` with `summarize()` for descriptives and **ggplot2** (`fill` or `color`) to compare groups with **histograms**, **dot plots**, or **boxplots**.

Two-sample t-test is optional — Project 3 is descriptive only.

Read 2 — Scatterplots and two-way tables

Essential · after Read 1 and Quiz A

- Make a **scatterplot** in **ggplot2** for two **numerical** variables (`aes(x, y)` and `geom_point()`).
- Describe an association using **form** (e.g. linear vs curved), **direction** (positive, negative, or none), and **strength** (strong, moderate, weak, or none).
- Build a **two-way table** of **counts** and compute **joint**, **row**, and **column proportions** (know which denominator you use).
- Draw **stacked bar charts** for counts and **100% stacked** bars (`position = "fill"`) to show **proportions within** bars.
- Implement tables and plots with `count()`, `group_by()`, `mutate()`, and `pivot_wider()`.

Legacy lecture pages remain linked from the 15-week index for reference.

Module 4 — Designing Fair Comparisons

Comparing two groups responsibly.

Course path

Step	Material
Read 0	Module 4 Read 0 — project preview & fair-comparison vocabulary
Read	Module 4 Read 1
Quiz A	Def & Notation
Read more	Module 4 Read 2
Quiz B	Concept Check
Demo lab	Module 4 demo
Exercise lab	Module 4 exercise lab
Milestone	Project 4 milestone *(Canvas checkpoint)*
Synthesis	Module 4 synthesis
Project	Project 4

Read 1 — Causal questions, confounding, and the RCT

Essential · after Read 0

- Compare **randomized controlled trials** and **observational** designs when the goal is a causal claim about a treatment or exposure.
- Explain how participants end up in comparison groups in an **RCT** versus an **observational study**.
- State what a *p*-value does and does **not** address for a difference between two groups.

- Define a **confounder** in words and recognize plausible confounders in a stated observational comparison.
- Explain how a confounder can **create or distort** an association between an explanatory variable and a response.
- Describe **subclassification / stratification** when a confounder is measured.

Read 2 — Two-group inference from the methods table

Essential · after Read 1 and Quiz A

- Use the **Methods table** to pick `t.test()` for two independent numerical groups or `prop.test()` / `chisq.test()` for binary comparisons, and name key conditions (e.g. independence, expected counts).
- Interpret **Welch two-sample *t*-test** output (hypotheses, p-value, confidence interval) in context.
- Connect **random assignment** in an RCT to reduced risk of confounding for treatment–outcome comparisons.

Legacy lecture pages remain linked from the 15-week index for reference.

Module 5 — Looking Ahead

Using data to make predictions.

Course path

Step	Material
Read 0	Module 5 Read 0 — project preview & four-day schedule
Read	Module 5 Read 1
Quiz A	Def & Notation
Read more	Module 5 Read 2
Quiz B	Concept Check
Demo lab	Module 5 demo
Exercise lab	Module 5 exercise lab
Synthesis	Module 5 synthesis
Milestone	Canvas Module 5 *(team, option, data plan)*
Project	Project 5 · **Portfolio package due Thu Jul 2**

Read 1 — Correlation and the least-squares line

Essential · after Read 0

- Read a scatterplot using **form** (linear or curved), **direction** (positive or negative), and **strength** (strong, moderate, or weak).
- Compute and interpret **Pearson correlation** r as a measure of strength and direction of a linear association.
- Fit a **simple linear regression** line $\hat{y} = b_0 + b_1 x$ with `lm()` and identify slope b_1 and intercept b_0.
- Interpret the **slope** in context: predicted change in y for a one-unit increase in x.
- Use the fitted equation to make a **point prediction** for a new x value.
- Define a **residual** as observed minus predicted $(y - \hat{y})$ and explain what the least-squares criterion minimizes.
- Interpret **MSE** and R^2 on a test set as summaries of predictive performance on new data.

Optional enrichment (not required for Project 5): Logistic regression — binary response with `glm()`

Read 2 — Inference in simple linear regression

Essential · after Read 1 and Quiz A

- State the residual-normality assumption that supports interval inference in linear regression.
- Interpret a **95% confidence interval for the slope** β_1 in context.

- Interpret a **confidence interval for the mean response** at a specific x_0 value.
- Interpret a **prediction interval** for an individual future response at x_0.
- Explain why **prediction intervals** are wider than confidence intervals for the mean response.
- Use `confint()` and `predict(..., interval = "prediction")` in R to extract these intervals.

Legacy lecture pages remain linked from the 15-week index for reference.

Not on the async course path

These pages remain from the 15-week course for reference. They are not assigned on the course path.

Lecture 28

Inference for two-group comparisons — causal questions and study design

- Compare **randomized controlled trials** and **observational** designs when the goal is a causal claim about a treatment or exposure.
- Explain how participants end up in comparison groups in an RCT versus an observational study.
- State what a p-value does and does **not** address for a difference between two groups.

Lecture 29

Confounding

- Define a **confounder** in words and recognize plausible confounders in a stated observational comparison.
- Explain how a confounder can create or distort an association between an explanatory variable and a response.
- Describe **subclassification / stratification** when a confounder is measured.

Lecture 30

Study design, p-values, and two-group inference

- Use the **Methods table** to pick **`t.test()`** for two independent numerical groups or **`prop.test()`** / **`chisq.test()`** for binary comparisons, and name key conditions.
- Interpret **Welch two-sample t-test** output (hypotheses, p-value, confidence interval) in context.
- Connect **random assignment** in an RCT to reduced risk of confounding for treatment–outcome comparisons.

Lecture 31

Matched pairs — differences, plots, paired t-test, and McNemar

- Identify a **matched pairs** design and compute **within-pair differences** (or use a paired test) when the same units contribute two measurements.
- Summarize paired numerical data with plots (e.g. boxplot of differences) and run a **paired t-test** (or one-sample test on differences).
- For **two binary outcomes on the same units**, build the right table and run **McNemar's test** when appropriate.

Lecture 32

One-way ANOVA — comparing means across more than two groups

- Recognize when the outcome is **numerical** and the explanatory factor has **more than two levels**, so a one-way **ANOVA** is the natural extension of a two-sample t-test.
- State the **omnibus** null hypothesis for equal population means across groups.

Lecture 33

Chi-square tests — independence and goodness of fit

- Build **contingency tables** of counts for two categorical variables and use **`chisq.test()`** as appropriate.
- Check **expected counts** (and know when **`simulate.p.value = TRUE`** is needed).
- Relate chi-square tests to the **"by chance"** explanation for a table.

Lecture 34

Work session — Project 4 (simulation, notebook, and writeup)

- Apply study-design ideas (RCT vs observational, confounding, DAG) to **your** assigned two-group question.
- Produce a **reproducible** simulated dataset (`set.seed`) and a notebook with **descriptives**, **plots**, and the **correct test/CI** from the Methods table.

Lecture 35

Correlation and simple linear regression

- Interpret a **correlation coefficient** (strength and direction) for two numerical variables.
- Fit and interpret a **simple linear regression** line as a model for how the mean of Y changes with x.
- Use **scatterplots** and diagnostic thinking about **linear** vs **curved** form.

Lecture 36

Linear regression inference — confidence and prediction intervals

- Interpret confidence intervals for the slope and for the mean response at a chosen value of x.
- Interpret prediction intervals for an individual future response and explain why they are wider than mean-response intervals.
- Connect residual assumptions to validity of interval-based inference.

Lecture 37

Logistic regression — modeling a binary outcome

- Recognize when the response is **binary** and a **logistic** (log-odds) model is appropriate instead of ordinary least squares regression.
- Interpret **fitted probabilities** or **odds ratios** from simple logistic output in context (at the level taught in this course).

Lecture 38

Ethics and statistical reasoning review

- Use a truth-first lens to connect design, assumptions, missingness, and communication.
- Evaluate ethical pitfalls in generalization, causal language, reporting, and model validity.
- Practice responsible statistical wording and evidence-based conclusions.